Analysis 2: HW6

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Problem 1.
First consider f' i.e the Jacobian of f. Now we know that this is a $n \times d$ matrix with rank r. So we can find r columns of the d columns of this matrix which are linearly independent and hence form a basis for some r dimensional subspace of R^n. So consider these coordinates (the ones that correspond to these r columns) beginning (we're permuting it toward the beginning) and the $n-r$ to the end such that we have $(x, y) \in R^{d-r} \times R^r$. Now note that the matrix also has r independent rows which correspond to the output of the function. So divide our function f into $f = (f_1, f_0)$ such that $f_1 : U \to R^r$ and $f_0 : U \to R^{n-r}$. Here f_1 corresponds to the r components and f_0 the $n-r$ components of our function. This gives us $\partial_y f_1(x, y)$ is an invertible matrix.

Now note that as $\partial_y f_1(x, y)$ is an invertible matrix, we can apply the implicit function theorem so locally we have a diffeomorphism i.e $\phi(x, y) = (x, f_1(x, y))$. Note that $\phi : R^d \to R^d$. Now let $x' = x$ and $y' = f_1(x, y)$. So we have $\phi(x, y) = (x', y')$ and we have $(x, y) = \phi^{-1}(x', y')$ because ϕ has an inverse. Now note that we get,

$$f \circ \phi^{-1}(x', y') = f(x, y) = (f_1(x, y), f_0(x, y)) = (y', f_0 \circ \phi^{-1}(x', y'))$$

Now let $h(x', y') = f_0(\phi^{-1}(x', y'))$ so we can rewrite the above as,

$$f \circ \phi^{-1}(x', y') = (y', h(x', y'))$$

so we have a map from $(x', y') \to (y', h(x', y'))$. Now this map has rank r because of y', so the jacobian is zero as if it was non-zero then that means we would have a higher rank derivative. So this means that h is a constant say c. This gives us,

$$f \circ \phi^{-1}(x', y') = (y', c)$$

Now we can just find a map ψ that centers c at 0. So define $\psi(a, b) = (a, b - c)$ so if we apply this to (y', c) we have $\psi(y', c) = (y', 0)$. Clearly ψ is a diffeomorphism. So we have,

$$\psi \circ f \circ \phi^{-1}(x', y') = (y', 0, \dots, 0)$$

which is what we want to show.
Problem 2. We have $E \subseteq R^d$ a closed set and $f : E \to E$ continuous such that,

$$|f(x) - f(y)| \leq \alpha |x - y|$$

f for some $\alpha \in (0, 1)$. We need to show that there is unique fixed point p such that $f(p) = p$. Consider an arbitrary point p_1 now define $p_{i+1} = f(p_i)$, so we have $p_2 = f(p_1)$ and so on. Our goal will be to show that the sequence (p_i) is a convergent sequence and converges to a unique limit say p.

Firstly applying the above condition we know that,

$$|f(p_i) - f(p_{i-1})| \leq \alpha |p_i - p_{i-1}|$$

but we have $f(p_i) = p_{i+1}$ so this give us,

$$|p_{i+1} - p_i| \leq \alpha |p_i - p_{i-1}|$$

Now clearly we can inductively continue this to get,

$$|p_{i+1}-p_i| \leq \alpha|p_i-p_{i-1}| \leq \alpha(\alpha|p_{i-1}-p_{i-2}|) \leq \cdots \leq \alpha^{i-1}|p_2-p_1|$$

But note that $p_2 = f(p_1)$ so we have,

$$|p_{i+1}-p_i| \leq \alpha^{i-1}|p_2-p_1| \leq \alpha^{i-1}|f(p_1)-p_1|$$

Now note that as we fixed p_1 we have $|f(p_1)-p_1|$ is just a constant say $c > 0$. So we have,

$$|p_{i+1}-p_i| \leq \alpha^{i-1}c$$

Now consider $|p_m - p_n|$ for $m > n$ we have,

$$\begin{aligned}
|p_m - p_{m-1} + p_{m-1} + \cdots - p_n| &\leq |p_m - p_{m-1}| + |p_{m-1} + p_{m-2}| + \cdots + |p_{n+1} - p_n| \\
&\leq \alpha^{m-2}|p_2-p_1| + \alpha^{m-3}|p_2-p_1| + \cdots + \alpha^{n-1}|p_2-p_1| \\
&\leq c(\alpha^{m-2} + \alpha^{m-3} + \cdots + \alpha^{n-1}) \\
&\leq c\alpha^{n-1}(\alpha^{m-n-1} + \cdots + 1) \\
&\leq c\alpha^{n-1}\frac{1-\alpha^{m-n}}{1-\alpha} \\
&\leq c\frac{\alpha^{n-1}}{1-\alpha}
\end{aligned}$$

Now for any $\varepsilon > 0$ consider n for which we have,

$$\begin{aligned}
c\frac{\alpha^{n-1}}{1-\alpha} &< \varepsilon \\
\alpha^{n-1} &< \frac{\varepsilon(1-\alpha)}{c} \\
(n-1)\log(\alpha) &< \log\left(\frac{\varepsilon(1-\alpha)}{c}\right) \\
n-1 &> \frac{\log\left(\frac{\varepsilon(1-\alpha)}{c}\right)}{\log(\alpha)} \qquad \text{note because } \log(\alpha) \text{ is negative} \\
n &> \frac{\log\left(\frac{\varepsilon(1-\alpha)}{c}\right)}{\log(\alpha)} + 1
\end{aligned}$$

so let $N_\varepsilon = \frac{\log\left(\frac{\varepsilon(1-\alpha)}{c}\right)}{\log(\alpha)} + 1$.

So we have for any $\varepsilon > 0$ there exists some N_ε such that if $m, n > N_\varepsilon$ we have,

$$|p_m - p_n| \leq c\frac{\alpha^{n-1}}{1-\alpha} < \varepsilon$$

which means that we have (p_i) is a Cauchy sequence and hence is a convergent sequence. Now this implies that we have $(p_i) \to p$ which is the limit of this sequence. Now note that we have for large enough n (chosen for say $\frac{\varepsilon}{2}$)

$$\begin{aligned}
|f(p)-p| &= |f(p) - f(p_n) + f(p_n) - p| \\
&\leq |f(p) - f(p_n)| + |f(p_n) - p| \\
&\leq \alpha|p_n - p| + |p_{n+1} - p| \\
&\leq \alpha\frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\
&\leq \frac{\varepsilon}{2}(1+\alpha) \leq \varepsilon
\end{aligned}$$

Or we can also just apply the limit to $p_{i+1} = f(p_n)$ to get $p = f(p)$.

Now note that we showed this existance for the sequence $p_1, p_2 \dots$. But we have yet to show that for any arbitrary starting point we converge to the same sequence. So consider the point q_1 and consider the sequence defined the same way as p_1, so $(q_1, f(q_1), \dots)$, we need to show the sequence converges to the same value p. Clearly using the same lines of reasoning we can establish a limit say $(q_i) \to q$. We need to show that $p = q$. First for $\varepsilon > 0$ choose N_1, N_2 (corresponding to $\frac{\varepsilon}{3}$) for each of our sequences such that we have,

$$|p_n - p| < \frac{\varepsilon}{3} \quad \text{and} \quad |q_n - q| < \frac{\varepsilon}{3} \text{ respectively for } n > N_1 \text{ and } n > N_2$$

Now choose the larger of these two i.e $n > N = \max\{N_1, N_2\}$ so that we have both the above,

Now note that we have for $i > N$ that

$$\begin{aligned} |p - q| &= |p - p_i + p_i - q_i + q_i - q| \\ &\leq |p - p_i| + |p_i - q_i| + |q_i - q| \end{aligned}$$

Now note that we can use our contraction to get,

$$\begin{aligned} |p_i - q_i| &\leq \alpha |p_{i-1} - q_{i-1} \\ &\leq \alpha^{i-1} |p_1 - q_1| \end{aligned}$$

now clearly $|p_1 - q_1|$ is a constant, so we can just make the above arbitrarily small i.e smaller than $\frac{\varepsilon}{3}$, putting this together we have,

$$|p - q| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

for any $\varepsilon > 0$ which means we have $p = q$.

Problem 3.

1. First we show that if p is a local minima then $f'(p) = 0$ and $f''(p) \geq 0$. Our idea will be similar to the 1-D but we consider an arbitrary direction first say v. Then we have $g(t) = f(p + tv)$ where t is some value. Now clearly the local minima of g is when we have $g(t_0) = f(p)$ as p is the local minima of f. So we have $t = 0$. But note that $g'(t) = \nabla f(p + tv)v$. But we know that $g'(0) = 0$ (just the single variable case) so this means that $\nabla f(p)v = 0$ for any direction v, this means that we have $\nabla(p) = 0$.

Now again note that we from the single variable case that $g''(0) \geq 0$, but if $g(t) = f(p + tv)$ then $g'(t) = f'(p + tv)v$ so $g''(t) = \frac{v^2}{2} f''(p + tv)$ we're putting v^T in the start to make the dimensions match when computing the matrix multiplication. So we know that $g''(0) \geq 0$ which means that $v^T f''(p)v \geq 0$ for all v, clearly this means that we need $f''(p) \geq 0$.

2. Now we need to show that $f'(p) = 0$ and $f''(p) > 0$ means that $p$is the local minima. We can use the Taylor expansion to get,

$$f(p + h) = f(p) + hf'(p) + \frac{h^2}{2} f''(p) + remainder$$

But we knwo that $f'(p) = 0$ and that $f''(p) \geq 0$. So,

$$f(p + h) = f(p) + \frac{h^2}{2} f''(p) + remainder$$

Now note that for small enough h we have the remainders goes to zero (and we can ensure that $\frac{h^2}{2} f''(p) + remainder$ is positive) so this gives us,

$$f(p + h) \geq f(p)$$

for any h close to 0 and hence we have p is a local minima.